

Supplementary Information

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For: *Almost every gene set reads higher in the index arm: a size-matched empirical null for small rare-tumour expression series, and what it leaves of the EWSR1::NR4A3 direct-target catalogue.*

This supplement contains the detailed methods and complete supporting tables for the main text. Section S1 defines probe mapping, score eligibility and circularity; S2–S3 contain the evidence catalogue and set scores; S4–S5 report activity and robustness analyses; S6 gives sequence-scan parameters; and S7–S8 preserve the search and genome-build evidence. Every table is generated from a committed artifact rather than re-typed from prose. The producers are listed in the main text's Data and code availability section. The pre-registered rule in S1.3 is retained verbatim as a historical protocol, including its reported limitations.

S1 · Method detail

S1.1 · Probe mapping and floors

Probes were mapped to gene symbols per platform. GPL6244: 20,230 of 28,459 probes carry a symbol, giving 18,694 distinct symbols. GPL3290 maps 27,203 of 43,008 spots through an EST-accession bridge to 14,932 distinct symbols. A gene that cannot be read may be missing from the bridge rather than the array. We record it as unread, not absent.

Scores are standardized within each array. We z-score each sample's values against its own probe distribution. A gene's score is the mean z-score of probes mapped to that gene; a gene-set score is the mean over its readable members. We require **three samples per group** for a contrast, and **four genes and 0.4 coverage** for a gene-set score. Coverage here is the fraction of the set that the platform can read. A set below either floor produces no score and is marked accordingly. Class A contains three genes, so it is never scored alone as a set.

S1.2 · The circularity grade

Whether GSE4303 is the Subramanian (2005) cohort was graded from the fetched GEO series record, never from sample counts: the record's title, summary, contributors and linked PubMed identifier were read verbatim.

A record naming PMID 15920699 or Subramanian is graded **circular**. A record naming neither is graded *not-clean*, because omission is not evidence of independence. An unreadable record is graded *unanswered*.

The record does name both, so the verdict is circular, which is what makes both the set-E score and the *PPARG* gene row on that platform non-independent (main text §3.8).

S1.3 · The pre-registered decision rule, and how it landed

A six-branch decision rule was written and committed while the measurement run was still executing, so the verdict could not be fitted to whatever came back. Each branch carried its sentence, its ceiling and its next step in advance.

#	outcome	what it licenses
A	<i>ENO3</i> reproduces and class A (or A+B) clears its null on both platforms and <i>PLAGL1</i> reads down	A positive, EMC-specific result, with the ceiling attached in the same paragraph.
B	<i>ENO3</i> reproduces, class A clears its null on one platform only	A single-platform observation, reported as one.
C	<i>ENO3</i> reproduces, nothing clears its null	Still a result: the published target set is not distinguishable from a size-matched random gene set.
D	<i>ENO3</i> does not reproduce	Report the instrument and stop. No biological sentence may be written.
E	<i>ENO3</i> reproduces but <i>PLAGL1</i> reads up	Every up row loses its strongest defence against the offset explanation.
F	Filion Table 1 clears its null but class A does not	The instrument reads EMC and the fusion-target set is the thing that is flat.

Outcome F came true, with a per-gene positive inside it that the rule did not anticipate. Two limits of the pre-registration are recorded rather than quietly rewritten. First, the branches were written over *set* scores and the measurement landed at the *gene* level, so a future version of this rule needs an explicit gene-level branch. Second, the rule did not anticipate that the per-gene result would itself be split by a later confound audit — *ENO3* surviving every stratification while *SEMA3C* reverses sign — so it had no branch for "the genes disagree with each other", which is what happened.

S2 · The complete evidence-typed catalogue

Class assignments are from the main text Table 1. The **verbatim sentence** each classification rests on is held in the machine-readable catalogue (`nr4a3_fusion_targets.py` → `LITERATURE_TARGETS`, emitted to `nr4a3-fusion-targets.json` → `evidence_table.rows[.verbatim]`) and is not reproduced here only for length; every row below carries its citation and the assay the classification rests on.

A published negative control accompanies the catalogue: *CALDI*, whose promoter was searched for NOR-1 response elements in the same experiment that found the *SMPX* site, and none were found (PMID 27181368). It controls the inference "this gene moved, therefore NR4A3 bound it" — not EMC biology.

TABLE S1 — the complete evidence-typed catalogue

gene	class	factor actually tested	assays	cell system	species	expected in EMC	citation
ENO3	A	TFG::NR4A3 (TFG-TEC)	EMSA, ChIP (endogenous promoter), luciferase reporter, two NGFI-B response element motifs upstream of the putative TSS, ChIP for histone H3 acetylation at the endogenous promoter	cultured cell lines over-expressing TFG-TEC (the t(3;9) EMC fusion variant)	human (human beta-enolase promoter)	UP	Kim AY, Lim B, Choi J, Kim J. The TFG-TEC oncoprotein induces transcriptional activation of the human beta-enolase gene via chromatin modification of the promoter region. <i>Mol Carcinog</i> 2016. PMID 26310886, doi 10.1002/mc.22384
PPARG	A	EWSR1::NR4A3, NR4A3 (native), NR4A3-deltaC (native truncated)	predicted perfect NBRE at -675 bp (5' AAAGGTCA 3'), band-shift (EMSA) with the fusion protein, 2.8 kb human PPARG isoform-1 promoter luciferase reporter, single-nucleotide NBRE mutant of that reporter	CFK2 fetal RAT chondrogenic cells, stable EWSR1/NR4A3 lines (et2, et16, et19) and transient transfection of wild-type CFK2; HUMAN PPARG promoter construct	rat (the promoter construct is human)	UP	Filion C, Motoi T, Olshen AB, et al. The EWSR1/NR4A3 fusion protein of extraskeletal myxoid chondrosarcoma activates the PPARG nuclear receptor gene. <i>J Pathol</i> 2009;217(1):83-93. PMID 18855877, PMCID PMC4429309, doi 10.1002/path.2445
SEMA3C	A	EWSR1::NR4A3, TAF15::NR4A3, NR4A3 (native)	in-silico NBRE-like site (MatInspector, GRCh38 chr7), chromatin affinity purification + target qPCR (ChAP-qPCR), Strep-tagged	tBJ/ER transformed HUMAN fibroblasts engineered to express Strep-tagged NR4A3, EWSR1-NR4A3 (E-N) or TAF15-NR4A3 (T-N)	human	UP	Brenca M, Stacchiotti S, Fassetta K, et al. NR4A3 fusion proteins trigger an axon guidance switch that marks the difference between EWSR1 and TAF15 translocated extraskeletal myxoid chondrosarcomas. <i>J Pathol</i> 2019;249(1):90-101. PMID 31020999, PMCID PMC6766969, doi 10.1002/path.5284
BIRC3	B	NR4A3 (native)	NBRE binding site	vascular smooth muscle cells / hypoxic endothelium	human/rodent vascular	UP	Reviewed in PMCID PMC6912296 and PMC8583700
CCND1	B	NR4A3 (native)	ChIP at the Cyclin D1 promoter, NBRE site	hepatocytes; vascular smooth muscle cells; guidewire arterial-injury model in NOR-1-deficient mice	mouse / rat vascular and hepatic cells	UP	Reviewed in Herring JA, Elison WS, Tessem JS. Function of Nr4a Orphan Nuclear Receptors in Proliferation, Apoptosis and Fuel Utilization Across Tissues. <i>Cells</i> 2019;8:1373. PMID 31683815, PMCID PMC6912296; and in Haller F, et al. <i>Nat Commun</i> 2019;10:368. PMID 30664630, PMCID PMC6341107
CDKN2AIP	B	NR4A3 (native)	ChIP at predicted sites in the CDKN2AIP promoter, luciferase reporter reversed by promoter mutant	MHCC-LM3 human hepatocellular carcinoma cells	human	UP	Zhao X, Min X, Wang Z, et al. NR4A3 inhibits the tumor progression of hepatocellular carcinoma by inducing cell cycle G0/G1 phase arrest and upregulation of CDKN2AIP expression. <i>Int J Biol Sci</i> 2024. PMID 39664575, PMCID PMC11628324, doi 10.7150/ijbs.95174

gene	class	factor actually tested	assays	cell system	species	expected in EMC	citation
COX5A	B	NR4A3 (native)	Cut&Tag over the promoter, dual-luciferase reporter	neonatal mouse cardiomyocytes; HEK293T reporter	mouse; human reporter construct	UP	Peng H, et al. eBioMedicine 2024;106:105268. PMID 39098108, PMCID PMC11334830
GLS2	B	NR4A3 (native)	ChIP-seq + mRNA-seq, dual-luciferase reporter, abolished by mutation of the predicted NR4A3 motif	Schwann cells (diabetic peripheral neuropathy model)	rat/mouse Schwann cells	UP	Pang B, Chen S, Bai Y, Zhang Y, Wang Z. NR4A3 alleviates diabetic neuropathy via GLS2-mediated mitochondrial repair and Schwann cell differentiation. iScience 2026. PMID 42028030, PMCID PMC13099357, doi 10.1016/j.isci.2026.115515
ICAM1	B	NR4A3 (native)	binding to the NBRE consensus site	TNF-stimulated endothelial cells / monocyte adhesion	human endothelial	UP	Reviewed in PMCID PMC8583700 / PMC10088923 / PMC9100886
LOXL2	B	NR4A3 (native)	named a direct NOR-1 target gene in the source review	cardiac fibroblast-to-myofibroblast switch, NOR-1 transgenic mice	mouse	UP	Reviewed in Martinez-Gonzalez J, et al. Int J Mol Sci 2021;22:11371, PMCID PMC8583700
MYH7	B	NR4A3 (native)	named a direct NOR-1 target gene in the source review	cardiac hypertrophy, NOR-1 transgenic mice	mouse	UP	Reviewed in PMCID PMC8583700
NOX1	B	NR4A3 (native)	gene silencing, luciferase reporter, site-directed mutagenesis, EMSA	vascular smooth muscle cells; co-localisation in human atheroma	human	UP	Reviewed in Martinez-Gonzalez J, et al. Int J Mol Sci 2021;22:11371, PMCID PMC8583700
PDP1	B	NR4A3 (native)	Cut&Tag over the promoter	neonatal mouse cardiomyocytes	mouse	UP	Peng H, et al. eBioMedicine 2024;106:105268. PMID 39098108, PMCID PMC11334830
SDHA	B	NR4A3 (native)	Cut&Tag, truncated-promoter dual-luciferase mapping to region R3 (predicted element AAAGTCAC)	neonatal mouse cardiomyocytes; HEK293T for the HUMAN SDHA promoter reporter	mouse cardiomyocytes; human HEK293T for the reporter	UP	Peng H, Yuan J, Wang Z, et al. NR4A3 prevents diabetes induced atrial cardiomyopathy by maintaining mitochondrial energy metabolism and reducing oxidative stress. eBioMedicine 2024;106:105268. PMID 39098108, PMCID PMC11334830, doi 10.1016/j.ebiom.2024.105268
SKP2	B	NR4A3 (native)	EMSA, ChIP, NBRE site in the SKP2 promoter	vascular smooth muscle cells	human/rodent VSMC (the review does not disambiguate)	UP	Reviewed in Martinez-Gonzalez J, et al. NR4A3: A Key Nuclear Receptor in Vascular Biology, Cardiovascular Remodeling, and Beyond. Int J Mol Sci 2021;22:11371. PMID 34768801, PMCID PMC8583700

gene	class	factor actually tested	assays	cell system	species	expected in EMC	citation
SMPX	B	NR4A3 (native)	promoter deletion, site-directed mutagenesis of a non-consensus NBRE (-167/-160), EMSA, ChIP in differentiating human skeletal myoblasts	human vascular smooth muscle cells and HSMM myoblasts	human	UP	Ferran B, Marti-Pamies I, Alonso J, et al. The nuclear receptor NOR-1 regulates the small muscle protein, X-linked (SMPX) and myotube differentiation. <i>Sci Rep</i> 2016;6:25944. PMID 27181368, PMCID PMC4867575
TH	B	NR4A3 (native)	transient transfection through an NBRE site in the TH promoter	vascular smooth muscle cells; NOR-1 transgenic mouse aorta	mouse / VSMC	UP	Reviewed in Martinez-Gonzalez J, et al. <i>Int J Mol Sci</i> 2021;22:11371, PMCID PMC8583700
VCAM1	B	NR4A3 (native)	binding to the NBRE consensus site	TNF-stimulated endothelial cells / monocyte adhesion	human endothelial	UP	Reviewed in Martinez-Gonzalez J, et al. <i>Int J Mol Sci</i> 2021;22:11371, PMCID PMC8583700; and PMCID PMC10088923
VTN	B	NR4A3 (native)	listed by an independent review as a functionally validated direct target, over-expression + blocking-antibody / silencing rescue of migration, co-localisation in human atherosclerotic lesions	vascular smooth muscle cells; independently raised >2-fold by NR4A3 over-expression in the human MHCC-LM3 hepatocellular line	human	UP	Haller F, et al. <i>Nat Commun</i> 2019;10:368. PMID 30664630, PMCID PMC6341107 (target list); Martinez-Gonzalez J, et al. <i>Int J Mol Sci</i> 2021;22:11371, PMCID PMC8583700 (VSMC); Zhao X, et al. <i>Int J Biol Sci</i> 2024, PMCID PMC11628324 (MHCC-LM3)
PLAGL1	C	EWSR1::NR4A3	differential display in a fusion-expressing line, RT-PCR in six EMC tumours	CFK2 chondrogenic cells over-expressing EWS/NOR1; human EMC tumours vs immortalised and primary human chondrocytes	rat cells; human tumours	DOWN	Filion C, et al. The PLAGL1 gene is down-regulated in human extraskeletal myxoid chondrosarcoma tumors. <i>Cancer Lett</i> 2005. PMID 16112421, doi 10.1016/j.canlet.2004.12.007
SGK1	C	EWSR1::NR4A3	differential display in a tetracycline-regulated fusion line, co-immunocytochemistry, immunohistochemistry in 10 fusion-positive EMC	CFK2 fetal RAT chondrogenic cells with tetracycline-controlled EWS/NOR1	rat	FLAT_OR_DOWN_AT_TRANSCRIPT_LEVEL	Labelle Y, et al. Serum- and glucocorticoid-regulated kinase 1 (SGK1) induction by the EWS/NOR1(NR4A3) fusion protein. <i>Biochem Biophys Res Commun</i> 2006. PMID 16756948, doi 10.1016/j.bbrc.2006.05.134
NDRG2	D	□	Affymetrix U133A microarray, 3 fusion-positive EMC vs 137 other sarcomas, Western blot, immunohistochemistry in 9/9 EWSR1/NR4A3-positive EMC	human EMC tumour tissue	human	UP	Filion C, et al. <i>J Pathol</i> 2009;217(1):83-93. PMID 18855877, PMCID PMC4429309

S3 · Every scored gene set, with its detectability threshold

The **threshold** is the delta needed to exceed the 95% band of the size-matched null. The **reached** column divides the observed delta by that threshold: values below 1 are fractions of the threshold, and values at or above 1 are multiples. The A+B target set reached 39% and 88% of its thresholds. The published EMC signature exceeded its thresholds 11.9× and 4.2×. The column shows how far each set lies from the stated criterion, rather than reporting only whether it passed.

Two results exceed their competitive thresholds only narrowly: the PPAR γ KO_UP falsifier and the TRRUST human-curated set on GPL6244 each reach 1.02×. Neither is a main-text conclusion. An excess of 2% should not be read as equivalent to an excess of 12-fold; the table reports the multiples to keep that distinction visible.

Set E is circular on GPL3290 and is reported for completeness only (§S1.2).

TABLE S2 — every scored gene set, with its detectability threshold

set	platform	n genes	Δ	null 95% band	threshold	reached	verdict
A_fusion_dna_binding_targets	GPL6244	3	—	—	—	—	no score (below floor)
A_fusion_dna_binding_targets	GPL3290	3	—	—	—	—	no score (below floor)
A_plus_B_all_dna_binding	GPL6244	19	+0.0403	[-0.1418, +0.1047]	0.10465	39%	not distinguishable
A_plus_B_all_dna_binding	GPL3290	17	+0.3301	[-0.2972, +0.3765]	0.37648	88%	not distinguishable
B_native_nr4a3_dna_binding_targets	GPL6244	16	-0.0675	[-0.1568, +0.1142]	-0.15684	43%	not distinguishable
B_native_nr4a3_dna_binding_targets	GPL3290	14	-0.1453	[-0.3357, +0.3971]	-0.33573	43%	not distinguishable
C_fusion_expression_only	GPL6244	2	—	—	—	—	no score (below floor)
C_fusion_expression_only	GPL3290	2	—	—	—	—	no score (below floor)
D_filion_table1_emc_vs_137_sarcomas	GPL6244	21	+1.1311	[-0.1374, +0.0947]	0.09472	11.94×	SET-SPECIFIC
D_filion_table1_emc_vs_137_sarcomas	GPL3290	18	+1.4783	[-0.2853, +0.3481]	0.34814	4.25×	SET-SPECIFIC
E_filion_table2_overlap_with_subramanian	GPL6244	20	+0.8932	[-0.1399, +0.0978]	0.0978	9.13×	SET-SPECIFIC
E_filion_table2_overlap_with_subramanian	GPL3290	18	+1.9850	[-0.2853, +0.3481]	0.34814	5.70×	SET-SPECIFIC
F_brenca_EWSR1_high_axon_guidance	GPL6244	3	—	—	—	—	no score (below floor)
F_brenca_EWSR1_high_axon_guidance	GPL3290	3	—	—	—	—	no score (below floor)
G_brenca_TAF15_high_axon_guidance	GPL6244	10	-0.4975	[-0.1957, +0.1460]	-0.19574	2.54×	SET-SPECIFIC
G_brenca_TAF15_high_axon_guidance	GPL3290	10	+0.1214	[-0.4059, +0.4600]	0.45998	26%	not distinguishable
PPARG_adipogenesis_process_proxy	GPL6244	189	+0.0473	[-0.0553, +0.0174]	0.01735	2.73×	SET-SPECIFIC
PPARG_adipogenesis_process_proxy	GPL3290	176	+0.2183	[-0.0665, +0.1325]	0.13251	1.65×	SET-SPECIFIC
PPARG_pparg_KO_DOWN	GPL6244	188	+0.0003	[-0.0548, +0.0173]	0.0173	2%	not distinguishable
PPARG_pparg_KO_DOWN	GPL3290	157	+0.2219	[-0.0715, +0.1383]	0.13828	1.60×	SET-SPECIFIC
PPARG_pparg_KO_UP_FALSIFIER	GPL6244	231	-0.0536	[-0.0527, +0.0143]	-0.05266	1.02×	SET-SPECIFIC
PPARG_pparg_KO_UP_FALSIFIER	GPL3290	196	-0.1118	[-0.0612, +0.1309]	-0.06123	1.83×	SET-SPECIFIC
PPARG_pparg_OE_UP	GPL6244	250	-0.0238	[-0.0507, +0.0122]	-0.05069	47%	not distinguishable
PPARG_pparg_OE_UP	GPL3290	230	-0.0020	[-0.0505, +0.1206]	-0.05054	4%	not distinguishable
PPARG_pparg_chip_chea	GPL6244	188	+0.0800	[-0.0548, +0.0173]	0.0173	4.62×	SET-SPECIFIC
PPARG_pparg_chip_chea	GPL3290	169	+0.2938	[-0.0636, +0.1336]	0.13357	2.20×	SET-SPECIFIC
PPARG_pparg_curated_trust_human	GPL6244	63	+0.0454	[-0.0870, +0.0445]	0.04448	1.02×	SET-SPECIFIC
PPARG_pparg_curated_trust_human	GPL3290	57	+0.1647	[-0.1374, +0.2115]	0.21148	78%	not distinguishable

S4 · The PPAR γ receptor-activity reading, and its adipogenic ceiling

PPARG abundance in EMC is settled elsewhere and is not this work's subject. What no study retrieved in the corpora searched here reports is receptor *activity* — transcriptional output, as distinct from receptor abundance.

This is a bounded statement about a search, not a claim that no such measurement exists anywhere.

Six gene sets were scored, each pinned to a verbatim Enrichr term with its species read off the term rather than assumed, each null-calibrated on its own platform.

arm	library / term	species	GPL6244	GPL3290
ChEA ChIP-PET targets (191)	ChEA_2022 · PPARG 19300518 ChIP-PET 3T3-L1 Mouse	mouse	+0.080, p_emp ≤ 0.0005 → SET-SPECIFIC UP	+0.294, p_emp ≤ 0.0005 → SET-SPECIFIC UP
KO_UP falsifier (246)	TF_Perturbations... · PPARG DEFICIENCY MOUSE GSE23421 ... UP	mouse	−0.054, p_emp 0.041 → SET-SPECIFIC DOWN	−0.112, p_emp 0.0035 → SET-SPECIFIC DOWN
KO_DOWN (206)	...PPARG DEFICIENCY MOUSE GSE23421 ... DOWN	mouse	+0.0003 → not distinguishable	+0.222, p_emp ≤ 0.0005 → SET-SPECIFIC UP
OE_UP (269)	...PPARG OE MOUSE GSE10192 ... UP	mouse	−0.024 → not distinguishable	−0.002 → not distinguishable
TRRUST, human-curated (66)	TRRUST...2019 · PPARG human	human	+0.0454, p_emp 0.048 → SET-SPECIFIC UP	+0.1647 → not distinguishable
adipogenesis process proxy (200)	MSigDB_Hallmark_2020 · Adipogenesis	unstated in the term	+0.047, p_emp ≤ 0.0005 → SET-SPECIFIC UP	+0.218, p_emp ≤ 0.0005 → SET-SPECIFIC UP

Why KO_DOWN and OE_UP cannot be expected to agree, measured rather than argued. The two arms were once read as disagreeing about biology, but they share 16 genes out of 206 and 269 (Jaccard 0.035) and come from different experiments in different tissues.

They are, for practical purposes, different gene sets, and asking them to agree was asking two nearly disjoint lists of mouse genes to score alike in human tumour tissue.

Arithmetic control: $KO_DOWN \cap KO_UP = 0$, exactly as the two arms of one knockout experiment must be.

What replicates, and how strongly. The occupancy-derived target set is set-specific up on both platforms and remains strongly significant under exact label permutation ($p = 0.00033$ and 0.00075).

The KO_UP falsifier is set-specific down on both platforms under the competitive null — but **it does not reach significance under the permutation test on either** ($p = 0.362$ and 0.296), and it clears its competitive threshold by only $1.02\times$ on GPL6244.

The falsifier half of the "two arms separating in opposite directions" pattern therefore rests on the competitive null alone, and is reported at that weight.

Three of the six arms (KO_DOWN, OE_UP, TRRUST) were never permutation-tested at all, so their verdicts rest on the competitive null only.

The ceiling is not small. The adipogenesis process proxy is also set-specific up on both platforms and shares 44 genes with the ChEA arm (23% of the smaller set), the largest overlap in the table, and it is itself significant under permutation on both platforms.

PPAR γ target output therefore cannot be separated from an adipogenic differentiation component in these data, and five of the six arms are mouse-derived.

Stated at full honesty: PPAR γ target genes are coordinately higher in EMC tumour tissue than in comparator sarcomas, beyond a size-matched random set, on two platforms — and the same data cannot distinguish that from an adipogenic differentiation programme.

This says nothing about the direction of any pharmacological intervention on this axis.

S5 · Robustness, stratification and sensitivity

The stratified table below is the one that separates the three class-A genes. Read down each gene's row: *ENO3* varies by no more than 0.011 SD across five dissimilar comparator sub-arms and is significant against each. *SEMA3C* changes from +1.657 against low-grade fibromyxoid sarcoma to −0.645 against desmoid fibromatosis. Both contrasts are significant, in opposite directions.

Note that the GPL3290 sub-arms have three comparators each, so **no p below $1/286 = 0.0035$ is reportable there however large the effect**; a printed value of 0.003497 represents that floor. It is not a finer measurement of significance.

The covariate-adjustment table carries its own internal control. The matrix panel separates the arms on GPL6244 ($\Delta -0.518$) and not on GPL3290 ($\Delta +0.006$); a covariate that does not differ between arms cannot move a contrast, and on GPL3290 none of the three genes moves by more than 3.5%.

That is the method behaving as it must, and it is the reason the GPL6244 movements can be read as adjustment rather than as noise.

The adjustment is a sensitivity analysis, not a correction. If a panel gene is itself driven by the fusion, the adjustment removes real signal. The provenance filter (main text §2.6) makes that unlikely but does not exclude it, and no adjusted number in this document is "the real effect".

TABLE S3 — robustness panel (exact permutation, BH, jackknife, rank re-read)

row	kind	platform	Δ	exact p	BH q	jackknife: sign holds in every fit	rank re-read: same sign (p)
ENO3	gene	GPL6244	+0.8074	7.3e-05	0.000438	yes	yes (4.9e-05)
PPARG	gene	GPL6244	+0.3071	0.0485	0.097	yes	yes (0.059668)
SEMA3C	gene	GPL6244	+0.7298	0.194266	0.233119	yes	yes (0.067107)
NR4A3	gene	GPL6244	+0.7415	0.000184	0.000552	yes	yes (0.000146)
PLAGL1	gene	GPL6244	-0.4234	0.082605	0.123908	yes	yes (0.015936)
SGK1	gene	GPL6244	-0.1807	0.369002	0.369002	yes	yes (0.445976)
A_plus_B_all_dna_binding	set	GPL6244	+0.0403	0.517577	—	yes	yes (0.36406)
B_native_nr4a3_dna_binding_targets	set	GPL6244	-0.0675	0.225747	—	yes	yes (0.137337)
D_filion_table1_emc_vs_137_sarcomas	set	GPL6244	+1.1311	1e-06	—	yes	yes (1e-06)
PPARG_pparg_chip_chea	set	GPL6244	+0.0800	0.000327	—	yes	yes (0.001238)
PPARG_pparg_KO_UP_FALSIFIER	set	GPL6244	-0.0536	0.361899	—	yes	yes (0.222558)
PPARG_adipogenesis_process_proxy	set	GPL6244	+0.0473	0.034511	—	yes	yes (0.037594)
ENO3	gene	GPL3290	+3.8113	0.000125	0.000625	yes	yes (0.000125)
PPARG	gene	GPL3290	+2.4809	0.000333	0.000833	yes	yes (0.000333)
SEMA3C	gene	GPL3290	+0.6228	0.16521	0.16521	yes	yes (0.097777)
PLAGL1	gene	GPL3290	-2.1340	0.002331	0.003885	yes	yes (0.008658)
SGK1	gene	GPL3290	+0.6156	0.155844	0.16521	yes	yes (0.257867)
A_plus_B_all_dna_binding	set	GPL3290	+0.3301	0.011114	—	yes	yes (0.001499)
B_native_nr4a3_dna_binding_targets	set	GPL3290	-0.1453	0.257493	—	yes	yes (0.334915)
D_filion_table1_emc_vs_137_sarcomas	set	GPL3290	+1.4783	0.0005	—	yes	yes (0.000375)
PPARG_pparg_chip_chea	set	GPL3290	+0.2938	0.000749	—	yes	yes (0.000375)
PPARG_pparg_KO_UP_FALSIFIER	set	GPL3290	-0.1118	0.296454	—	yes	yes (0.084416)
PPARG_adipogenesis_process_proxy	set	GPL3290	+0.2183	0.001499	—	yes	yes (0.00025)

TABLE S4 — every stratified comparator contrast, with its own exact permutation p

GPL6244 — comparator sub-arms: class_LGFMS_only (n=17), class_desmoid_fibromatosis_only (n=6), class_fibrosarcoma_only (n=6, whose GEO sample titles read *Myxofibrosarcoma*; the label is the scoring module's bucket name and the samples are myxoid), myxoid_comparators_only (n=23), non_myxoid_comparators_only (n=6)

gene	class LGFMS only	class desmoid fibromatosis only	class fibrosarcoma only	myxoid comparators only	non myxoid comparators only
<i>ENO3</i>	+0.805 (p 0.000168)	+0.807 (p 0.02165)	+0.816 (p 0.01082)	+0.808 (p 8e-05)	+0.807 (p 0.02165)
<i>PPARG</i>	+0.197 (p 0.2204)	+0.473 (p 0.04329)	+0.454 (p 0.06926)	+0.264 (p 0.1003)	+0.473 (p 0.04329)
<i>SEMA3C</i>	+1.657 (p 0.000119)	-0.645 (p 0.01515)	-0.523 (p 0.1364)	+1.089 (p 0.04609)	-0.645 (p 0.01515)
<i>NR4A3</i>	+0.839 (p 0.000158)	+0.837 (p 0.002165)	+0.370 (p 0.1255)	+0.717 (p 0.000568)	+0.837 (p 0.002165)
<i>PLAGL1</i>	-0.169 (p 0.4306)	-0.733 (p 0.04329)	-0.835 (p 0.06277)	-0.343 (p 0.1831)	-0.733 (p 0.04329)
<i>SGK1</i>	-0.032 (p 0.8796)	-0.167 (p 0.5173)	-0.617 (p 0.04762)	-0.184 (p 0.4055)	-0.167 (p 0.5173)

GPL3290 — comparator sub-arms: class_DFSP_only (n=3), class_GIST_only (n=3), non_myxoid_comparators_only (n=6), reference_pool_matched_only (n=3)

gene	class DFSP only	class GIST only	non myxoid comparators only	reference pool matched only
<i>ENO3</i>	+3.515 (p 0.003497)	+4.107 (p 0.003497)	+3.811 (p 0.000125)	+3.515 (p 0.003497)
<i>PPARG</i>	+2.679 (p 0.003497)	—	+2.481 (p 0.000333)	+2.679 (p 0.003497)
<i>SEMA3C</i>	+0.113 (p 0.8427)	+1.133 (p 0.05944)	+0.623 (p 0.1652)	+0.113 (p 0.8427)
<i>NR4A3</i>	—	—	—	—
<i>PLAGL1</i>	-1.659 (p 0.04242)	-2.609 (p 0.006061)	-2.134 (p 0.002331)	-1.659 (p 0.04242)
<i>SGK1</i>	-0.198 (p 0.5804)	+1.429 (p 0.01399)	+0.616 (p 0.1558)	-0.198 (p 0.5804)

TABLE S5 — covariate-adjusted sensitivity

platform	panel	arm separation Δ	gene	raw Δ	adjusted Δ	fraction retained	r(gene, proxy)
GPL6244	matrix (11)	-0.5184	<i>ENO3</i>	+0.8074	+0.6083	0.753	-0.368
GPL6244	matrix (11)	-0.5184	<i>PPARG</i>	+0.3071	+0.0992	0.323	-0.4798
GPL6244	matrix (11)	-0.5184	<i>SEMA3C</i>	+0.7298	+1.2458	1.707	0.3354
GPL6244	vascular (3)	-0.4057	<i>ENO3</i>	+0.8074	+0.6294	0.78	-0.3236
GPL6244	vascular (3)	-0.4057	<i>PPARG</i>	+0.3071	+0.0917	0.299	-0.489
GPL6244	vascular (3)	-0.4057	<i>SEMA3C</i>	+0.7298	+1.3789	1.889	0.4152
GPL3290	matrix (11)	+0.0064	<i>ENO3</i>	+3.8113	+3.8123	1.0	-0.036
GPL3290	matrix (11)	+0.0064	<i>PPARG</i>	+2.4809	+2.3939	0.965	-0.2575
GPL3290	matrix (11)	+0.0064	<i>SEMA3C</i>	+0.6228	+0.6148	0.987	0.6605
GPL3290	vascular (3)	-0.7962	<i>ENO3</i>	+3.8113	+2.2190	0.582	-0.5888
GPL3290	vascular (3)	-0.7962	<i>PPARG</i>	+2.4809	+1.0323	0.416	-0.7026
GPL3290	vascular (3)	-0.7962	<i>SEMA3C</i>	+0.6228	+0.5283	0.848	-0.0809

TABLE S6 — the skeletal-muscle admixture control (GPL6244)

gene	muscle-marker?	mean percentile in the 2 pooled muscle samples	EMC	comparators	EMC – comparator
<i>ACTA1</i>	yes	1.000	0.721	0.778	-0.057
<i>MYH7</i>	yes	1.000	0.479	0.522	-0.043
<i>PYGM</i>	yes	0.999	0.604	0.462	+0.142
<i>MYL1</i>	yes	0.998	0.133	0.283	-0.150
<i>ENO3</i>	no (class-A target)	0.996	0.663	0.348	+0.315
<i>SEMA3C</i>	no (class-A target)	0.924	0.945	0.685	+0.260
<i>PPARG</i>	no (class-A target)	0.381	0.707	0.591	+0.116

S6 · The NBRE motif scan — full parameters

Window. -10 kb/+15 kb around each gene's Ensembl-canonical TSS. The window is asymmetric, was fixed in advance on published regulatory architecture, and was frozen **before any sequence was read** rather than chosen around a result. A distal element outside it is untested by construction.

Motifs. The exact NBRE, 5'-AAAGGTCA-3' (PMID 1902986), and the NurRE (PMID 9315667), scanned on both strands with hits de-duplicated by genomic position so a palindromic match cannot be counted twice. **No NurRE occurs in any focus window.**

Two independent nulls, because an 8-mer occurs in 25 kb by chance. (i) A **dinucleotide-preserving shuffle** of the same window — 2,000 shuffles, seeded — which holds base and dinucleotide composition, and therefore GC and CpG content, exactly.

(ii) A **background panel** of 198 other genes' windows, read raw and additionally restricted to a GC-matched subset (± 0.05).

The panel is a gene list this project had already assembled for an unrelated question, so it cannot have been chosen to flatter or damage any gene here.

It averages **1.15 exact NBREs per 25 kb window**, so a single hit is what an arbitrary window contains anyway.

Sequences are cached, so the scan re-derives offline.

gene	exact NBRE	shuffle-null p	background-panel p	GC-matched p
<i>ENO3</i>	4	0.034	0.025	0.018
<i>PPARG</i>	3	0.227	0.131	0.209
<i>SEMA3C</i>	0	1.00	1.00	1.00
<i>NR4A3</i>	0	1.00	1.00	1.00
<i>NR4A1</i>	0	1.00	1.00	1.00

The one-mismatch (NBRE-like) class. Brenca *et al.* report a predicted NBRE-like site at *SEMA3C*, which by construction is not an exact NBRE, so the exact scan is not a test of their claim.

Octamers matching the NBRE with **at most one substitution** were therefore scanned with their own null, taken from the *same* shuffled sequences so every exact figure is unchanged. *SEMA3C*'s window carries **39** one-mismatch sites, the most among the three class-A genes (*PPARG* 32, *ENO3* 28). It is not the highest count among all windows: 7 of the 198 background windows have at least 39, with a maximum of 43.

The count does not reject the composition-based explanation: the null mean is 33.7, with $p = 0.203$ against the dinucleotide-preserving shuffle and $p = 0.118$ in the GC-matched comparison. These non-significant results do not prove that composition explains the count. Only the raw panel rank, which does not account for composition, suggests enrichment ($p = 0.040$). *SEMA3C*'s window is the most AT-rich of the focus set (GC 0.371), while the NBRE is itself AT-leaning; GC-matching addresses that possible source of a high raw rank.

The converse holds for *ENO3*: its one-mismatch count is *not* enriched (28 against a null mean of 25.6, $p = 0.336$), so its signal sits in exact NBREs rather than in degenerate ones.

A one-mismatch match is a weaker claim than an exact NBRE, not a stronger one: it admits many sequences per position, most of which no NR4A protein has been shown to bind. The calibration bounds a count; it does not license calling any hit a site.

The hit positions do not reproduce the published coordinates.

Kim *et al.* report two NBREs upstream of the *ENO3* TSS; this scan finds one upstream (−8.7 kb) and three downstream.

Filion *et al.* report a perfect NBRE at −675 bp of *PPARG*; the nearest hit here is −186.

Both papers numbered from their own promoter constructs, and *PPARG* has multiple promoters, so the offsets are not directly comparable.

This scan neither confirms nor refutes the published site positions, and a motif scan cannot refute a measured binding event at a named locus in any case.

S7 · The cohort search — every query, and every record it returned

Main-text §2.7 gives the method and §3.13 Table 10 the result. This section carries the queries verbatim, because a search reported only by its conclusion cannot be checked or repeated.

Table S7. The six queries, with the reason each was included.

#	query (NCBI E-utilities, db=gds)	why	as written	repaired
1	"extraskkeletal myxoid chondrosarcoma"[All Fields]	the disease name in full — highest precision	17	—
2	"myxoid chondrosarcoma"[All Fields] AND "expression profiling"[Filter]	the shortened name, restricted to expression series	0 $\triangle$	2
3	(EWSR1 AND NR4A3) OR "EWS-NOR1" OR "EWSR1-NR4A3" OR "TAF15-NR4A3"	the fusion rather than the disease — catches a deposit indexed by its driver	5	—
4	NR4A3[All Fields] AND sarcoma[All Fields] AND "expression profiling"[Filter]	the 3' partner plus lineage, for a deposit naming <i>NR4A3</i> but not EMC	0	0 — a confirmed zero
5	"chondrosarcoma"[All Fields] AND "expression profiling"[Filter] AND "Homo sapiens"[Organism]	deliberately over-broad: EMC samples inside a general chondrosarcoma series	0 $\triangle$	4
6	sarcoma[All Fields] AND "translocation"[All Fields] AND "expression profiling"[Filter]	translocation-sarcoma panels, the kind of deposit EMC hides inside	0 $\triangle$	32

All six executed and none returned an error. The counts are each query's own raw return and sum to 60; the 56 records in §3.13 are what remains after de-duplication across the six, so a deposit returned by two queries is counted once there and twice here.

Four queries first returned exactly zero, and that was a reading about the queries before it was a reading about GEO.

All four carry an "expression profiling"[Filter] clause; the two that returned records do not.

Query 5 asks GEO for human chondrosarcoma expression series — a question it cannot honestly answer with nothing.

A zero and a malformed query are the same length, and reported as "six queries, no fourth cohort" this would have made the negative sound materially stronger than the evidence supported.

The repair, and what it changed. A zero-returning query carrying a field restriction is re-asked with the restriction lifted and the search terms unchanged; records from the repaired form are used and the query is recorded as `read_after_syntax_repair`.

Three of the four then returned records — **2, 4 and 32** — taking the search from 7 series to **22** and from 22 records to **56**.

Query 6, the translocation-sarcoma panel query written precisely because it is the kind of deposit EMC hides inside, accounted for 32 of them on its own.

Query 4 returned zero again under the unrestricted re-ask and is the only zero in this table read as an absence.

Both forms and both counts are in `emc-cohort-search-inputs.json`.

The negative reported in Table 10 is the repaired search. The unrepaired one would have rested on two queries while presenting itself as six, and the seventeen sample-level zeros that give the negative its weight would not have existed — fifteen of those deposits arrived through the repaired queries.

What the search declines to conclude, and one way it reaches further than expected. GEO's `esearch` matches the text of a record, so a deposit that names the disease nowhere in its GEO metadata is invisible to every query above however many EMC samples it holds.

That bound is real and is the reason this is a statement about reach rather than existence.

But the index reaches past the series title and summary, which the returned records demonstrate directly.

Neither GSE43632 (*Large scale screening for fusion genes in sarcoma patient samples*) nor GSE80126 names EMC in its title or summary. Query 1 nevertheless returned both when it searched the full disease name under `[All Fields]`. The `[All Fields]` index therefore reaches text the module did not capture. Queries 4–6 were intended to find EMC samples within generically titled pan-sarcoma deposits. Their malformed form did not provide that coverage.

Both were then read at sample level and carry no EMC sample.

The coverage the broken queries were meant to provide was, in this instance, supplied by query 1 — which is a fortunate accident and not a design, and is recorded as such.

Records excluded for not being deposits. `db=gds` returns sample and platform records alongside series, and most of what came back was neither.

Of the 56 records: **21 series**, 1 curated dataset (GDS3481), **30 individual sample records** and **4 platform records**.

Ten of the 30 samples are samples of the three cohorts already analysed here — six EMC samples of GSE24369 and the four of the Brunner deposit, returned in their own right by the disease-name query — and two are the doxycycline-treated and untreated HEK293 samples of GSE11185.

A single sample is not a cohort and a platform is not a deposit; grading them as though they might be would bury the records that need a decision under records that never could.

Reproduction. `research/modalities/emc_cohort_search.py` → `emc-cohort-search.json`, with the raw query record in `emc-cohort-search-inputs.json`. The derive half runs offline from the cached queries and `--check` re-derives the verdict and diffs it against the committed artifact.

S8 · The Haller NR4A3 cistrome — how its genome build was determined

Main-text §3.11 and Table 9 report four deep NR4A3 peak sets from Haller *et al.* (Zenodo 10.5281/zenodo.1483691). This section records how they became usable, because the step is not routine and a reader should be able to reject it.

The problem. A BED file carries no genome build.

The deposit names none in its title, its description or its filenames, and the two candidate human builds differ by a large, position-dependent offset — on chr10, roughly 300 kb near *RET*.

An intersection performed on the wrong build does not fail: it returns a plausible number for a different locus.

This analysis therefore refuses to intersect any peak set whose build is unknown, and these four arrived in exactly that state.

The test. H3K4me3 marks active promoters.

The deposit includes an H3K4me3 peak set for each of the four samples, so on the correct build those peaks must recover most of a promoter panel — and on the wrong build they must not.

The recorded build-inference panel has 213 genes on hg19 and 211 on hg38, as listed in `emc-ret-cistrome-inputs.json` under `build_inference.ZENODO1483691.per_build`. These are the denominators for Table S8's recovery percentages. They differ from the occupancy-calibration panels in Table 9 (200 genes on hg19 and 198 on hg38) and from the separate 198-window motif background in §S6. The analyses use existing gene-list provenance, but their resolved panels and denominators must not be treated as identical.

Table S8. Promoter recovery of each H3K4me3 peak set, by candidate build.

sample	peaks	panel promoters recovered, hg19	hg38
AciCC-1	30,660	90.6%	33.6%
AciCC-2	30,532	92.5%	32.2%
AciCC-3	29,494	92.5%	33.2%
Normal parotid gland	28,407	93.9%	32.7%

The deposit is read as **hg19**, on a worst-case recovery of 0.906 against 0.322, a ratio of 2.81.

Why two thresholds and not one. A build is assigned only if the worst sample clears an absolute floor (0.80) *and* beats the runner-up by a ratio (2.0).

The ratio alone would accept two equally wrong builds; the absolute alone would accept a build on which peaks happen to be broadly distributed. **The 33% on hg38 is not noise** — the two builds are identical over much of the genome, so a substantial wrong-build recovery is expected, and a test that treated "it found some" as agreement would pass on either.

What the reader can check, and what would overturn it. The four biological samples agree on the build; an error confined to one peak file would not account for all four readings. Agreement across samples cannot exclude a shared annotation error. We used the mark with the strongest prior expectation of promoter localisation. We did not use a transcription factor, because a factor that genuinely avoided promoters could mimic the wrong build. Build inference and occupancy calibration both rely on gene annotations and the assigned coordinates, despite using different resolved panels. A shared error could therefore affect both steps; identical panel membership or denominators are not assumed.

If the deposit is in fact hg38, every Table 9 row from it is void — the class-A counts would refer to other loci — while the ChIP-Atlas and ReMap rows, whose builds come from their catalogues, are unaffected.

Nothing else in the manuscript depends on it.

Reproduction. `python3 research/modalities/emc_ret_cistrome.py --infer-builds`, offline, from the committed peak cache; the full per-build table is written to `emc-ret-cistrome-inputs.json` under `build_inference`.